

Project 1 — E-Commerce Sales Intelligence Dashboard

DAY 3

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Objective

Build a complete Exploratory Data Analysis (EDA) script producing 15 annotated business charts across 4 sections — Revenue, Delivery, Customer, and Advanced Analysis. Each chart includes a business insight in the title, making the output immediately readable by non-technical stakeholders and professional enough for a GitHub portfolio.

Key business metrics discovered

Metric	Value	Business implication
Total Revenue	R\$16.01M	Healthy GMV across 2-year period
Total Orders	99,441	Large enough dataset for reliable trend analysis
Avg Order Value	R\$160.99	Mid-range positioning — price-sensitive market
Avg Delivery Days	12.5 days	Significant logistics improvement opportunity
Late Order Rate	7.9%	~7,900 orders arrived after promised date
Avg Review Score	4.09 / 5	Strong satisfaction — product-market fit confirmed
Unique Customers	99,441	Nearly 1:1 ratio with orders — 97% one-time buyers
Unique Categories	71	Broad catalogue — concentration in top 10 is high

All 15 charts — descriptions and insights

#	Chart	Section	Key business insight
01	Monthly Revenue Trend	Revenue	Revenue peaks in November 2017 — Black Friday effect with a sharp MoM spike. Steady growth from mid-2017 onward shows platform maturity.
02	Top 10 Categories by Revenue	Revenue	Health & Beauty, Watches & Gifts, and Bed/Bath/Table together account for ~30% of total GMV. Category concentration is moderate.
03	Average Order Value by Month	Revenue	AOV is stable at ~R\$160 across the dataset period with minor promotional spikes. Suggests limited upsell/cross-sell activity.
04	Revenue by Day of Week	Revenue	Monday and Tuesday are the highest revenue days. Weekends underperform by ~18% — opportunity for targeted weekend promotions.
05	Revenue and Volume by Price Bucket	Revenue	Mid-range (R\$50-150) drives the most orders by volume. Premium (R\$150-500) drives highest total revenue — two distinct buyer segments.

06	Avg Delivery Days by State	Delivery	Northern states (RR, AP, AM) wait 2-3x longer than Sao Paulo. Clear logistics infrastructure gap in remote regions.
07	Late vs On-Time Orders	Delivery	7.9% of orders arrive late. Most deliveries complete within 10-20 days. Distribution is right-skewed with long tail for remote states.
08	Monthly Late Order Rate	Delivery	Late delivery rate spikes in early 2018 — correlates with rapid order volume growth outpacing logistics capacity at the time.
09	Freight Cost as % of Product Price	Delivery	Office furniture and large household items have freight > 40% of price — a significant margin risk for sellers in these categories.
10	Repeat vs One-Time Buyers	Customer / Advanced	97% of customers placed only 1 order. This is the single biggest business finding — severe retention problem and a massive growth lever.
11	Review Score Distribution	Customer / Advanced	Over 57% of rated orders received 5 stars — strong product-market fit. 1-star orders are worth analysing for common failure patterns.
12	Payment Type Split	Customer / Advanced	Credit card dominates at ~74% of orders. High instalment usage signals price sensitivity — buyers prefer to spread payments.
13	Revenue by State (Top 10)	Customer / Advanced	Sao Paulo alone contributes a dominant share of top-10 state revenue. Heavy geographic concentration — expansion opportunity in other states.
14	Order Status Breakdown	Customer / Advanced	Over 96% of orders successfully delivered. Cancelled and unavailable orders are worth monitoring for revenue leakage.
15	Revenue Heatmap: Month x Day of Week	Revenue	November weekdays are the hottest cells in the heatmap. Black Friday drives concentrated mid-week spend — not just a single-day event.

Challenges faced and fixes applied

#	Challenge	Root cause	Fix applied
1	TypeError: float object cannot be interpreted as integer (Chart 10)	repeat.max() returned NaN because some order_count values were NaN after the groupby. Passing NaN to range() caused the TypeError.	Added .dropna().astype(int) before calling max(). Also added a len(repeat) > 0 guard before calling ax2.hist() to prevent empty-series errors.
2	SyntaxError: unterminated string literal (Charts 11-15 one-liner attempt)	Tried to run Charts 11-15 as a PowerShell one-liner using python -c. PowerShell misinterpreted escape sequences (\n, \M) inside the string as PS escape sequences, breaking the string literal.	Abandoned the one-liner approach. Wrote charts_10_to_15.py as a proper script file. Downloaded and ran python notebooks/charts_10_to_15.py — all 6 charts saved cleanly.
3	FutureWarning on fillna inplace (carried over from Day 2)	pandas 2.x deprecates chained inplace operations. Warnings appear on every run.	Documented for refactor. Does not affect chart output. Will fix in next iteration using df[col] = df[col].fillna(value) pattern.

Files produced today

File path	Description
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notebooks/01_eda.py	Main EDA script — Sections 1-2 (Charts 1-9)
notebooks/charts_10_to_15.py	Fix script — Section 3-4 (Charts 10-15)
notebooks/charts/01_monthly_revenue.png	Chart 1 — Monthly Revenue Trend
notebooks/charts/02_top_categories.png	Chart 2 — Top 10 Categories
notebooks/charts/03_aov_monthly.png	Chart 3 — Avg Order Value
notebooks/charts/04_revenue_dow.png	Chart 4 — Revenue by Day of Week
notebooks/charts/05_price_bucket.png	Chart 5 — Price Bucket Analysis
notebooks/charts/06_delivery_by_state.png	Chart 6 — Delivery Days by State
notebooks/charts/07_late_orders.png	Chart 7 — Late vs On-Time
notebooks/charts/08_monthly_late_rate.png	Chart 8 — Monthly Late Rate
notebooks/charts/09_freight_pct.png	Chart 9 — Freight % of Price
notebooks/charts/10_repeat_buyers.png	Chart 10 — Repeat Buyers
notebooks/charts/11_review_scores.png	Chart 11 — Review Scores
notebooks/charts/12_payment_types.png	Chart 12 — Payment Types
notebooks/charts/13_revenue_by_state.png	Chart 13 — Revenue by State
notebooks/charts/14_order_status.png	Chart 14 — Order Status
notebooks/charts/15_heatmap.png	Chart 15 — Revenue Heatmap

Key learnings from Day 3

1. Always call `.dropna()` before `.astype(int)` on aggregated columns — `groupby` can produce `NaN` for missing groups.
2. PowerShell cannot handle complex python `-c` one-liners with escape sequences — always write a `.py` file instead.
3. Chart titles with embedded business insights (real numbers) make a portfolio project look professional vs generic.
4. `matplotlib hist()` needs integer bin boundaries — float max values from pandas will silently break the `range()`.
5. EDA annotations (markdown cells or title strings with actual numbers) are what hiring managers look for in notebooks.
6. 15 charts across 4 analytical sections (Revenue, Delivery, Customer, Advanced) covers all dimensions a BI interview could ask about.

Day 4 — next steps

Task	Detail
Install DuckDB	<code>pip install duckdb</code> — already done on Day 1

Load processed CSVs into DuckDB	<code>con.execute("CREATE TABLE fact AS SELECT * FROM 'data/processed/fact_orders.csv'")</code>
Write rolling 30-day revenue query	<code>SUM() OVER (ORDER BY date ROWS 29 PRECEDING)</code>
Write cohort analysis query	Group customers by first-purchase month, count repeat purchases
Write RFM query	Recency (days since last order), Frequency (order count), Monetary (total spend) per customer
Save all queries	<code>sql/01_rolling_revenue.sql, sql/02_cohort.sql, sql/03_rfm.sql</code>